

Customer Churn Analysis

Key Insights & Action Plan | End-to-End Data Analytics Project

A concise, evidence-based summary of the completed churn analysis across customer, subscription and support data.

Dataset 21 customers	Source tables Customer Subscription Support	Analysis focus Retention Churn Value Support
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Executive dashboard



Headline finding

The clearest pattern in this dataset is contract structure: monthly customers churned at 55.56%, versus 8.33% for annual customers. The high-risk segment also captured all six observed churners.

Important: this is a 21-customer portfolio dataset. Group-level percentages - especially state and support findings - should be treated as observed signals and validated on a larger sample.

Key insights

These findings translate the notebook results into business-facing statements: quantified where possible, comparative where useful, and cautious where sample sizes are small.

01 Contract structure drives the clearest churn gap

Monthly-contract customers churned at 55.56% (5/9), versus 8.33% (1/12) for annual contracts - approximately 6.7x higher. Five of the six churned customers were on monthly contracts.

02 Basic has the highest churn rate

Basic customers had a 60% churn rate (3/5), compared with 22.22% Standard (2/9) and 14.29% Premium (1/7). Basic customers accounted for 50% of all churned customers.

03 Customer count and customer value tell different stories

Although Basic had the highest churn rate, the single churned Premium customer contributed 1,150 of 2,047 churned CLTV - about 56% of the total churned CLTV.

04 Referral acquisition shows a strong observed churn signal

Referral customers churned at 83.33% (5/6), versus 16.67% Paid and 0% Organic. However, all five churned Referral customers were monthly-contract customers, so acquisition source and contract type should be considered together.

05 Churn peaked in September 2024

September 2024 recorded 2 of the 6 churn events (33.33%), the highest monthly count in the observed cancellation data. Other observed churn months recorded one event each.

06 Churn is concentrated in four observed locations

All six churned customers came from Karnataka, Meghalaya, Telangana and Delhi. Karnataka recorded 100% (2/2) and Meghalaya 66.67% (2/3); these rates are based on very small groups.

07 Competitor switching is the most common stated reason

Among the six churn cases, 2 (33.33%) cited switching to a competitor. Pricing, content dissatisfaction, streaming quality and forgotten trial cancellation each represented 16.67%.

08 Support activity is tightly linked to observed churn

Of the 7 customers with recorded support activity, 6 churned. All four recorded escalations were associated with churned customers, while escalation-vs-churn correlation was 0.47. The support sample is small, so this is an observed association, not causation.

The rule-based High Risk segment (churn score > 70) contained 6 customers, and all 6 churned. The Medium segment had 2 customers and 0 churns; the Low segment had 13 customers and 0 churns. This makes High Risk the clearest group for follow-up analysis, but the dataset is too small to establish predictive performance.

Action plan

The actions below are direct follow-ups to the observed patterns. They are framed as investigations and retention opportunities rather than unsupported conclusions.

01

Investigate the monthly-contract churn gap

Why: Monthly churn is 55.56% versus 8.33% annual, and 5 of 6 churned customers are monthly.

Next step: Review monthly vs annual pricing, renewal behavior, tenure, cancellation timing and customer experience; test whether contract migration is associated with lower churn on a larger sample.

02

Examine the Referral + Monthly segment

Why: Five of six Referral customers churned, and every churned Referral customer was on a monthly contract.

Next step: Break this segment down by plan, pricing, tenure, support activity and cancellation reason before attributing churn to the acquisition channel itself.

03

Investigate the churn concentration by location

Why: All six churns came from four observed locations, with Karnataka at 100% (2/2) and Meghalaya at 66.67% (2/3).

Next step: Check for location-specific pricing, service quality, product availability, support issues or acquisition mix - then validate against a larger customer base.

04

Review competitor and experience-related cancellations

Why: Competitor switching is the largest single cancellation reason at 33.33%; pricing, content, streaming quality and trial cancellation also appear.

Next step: Compare competitor value propositions and review complaints/support records around churn dates to identify recurring experience gaps.

05

Prioritize high-risk, high-value customers

Why: All six observed churners fall into the High Risk band, while one Premium churning alone represents about 56% of churned CLTV.

Next step: Use High Risk as a screening layer, then combine risk with CLTV and support signals to focus retention analysis on the customers with the greatest potential value at risk.

RETENTION LENS

Protect recurring value, not just customer count. The dataset shows why churn should be viewed through both volume and value: Basic drives more churn events, while Premium contributes a disproportionate share of churned CLTV.

Executive summary

The completed churn analysis shows an overall churn rate of 28.57% across the 21-customer dataset. The strongest observed pattern is contract structure: monthly customers churned at 55.56% compared with 8.33% for annual customers. Basic customers recorded the highest plan-level churn at 60%, while the Referral channel showed 83.33% churn. Churned customers account for 73.94 of monthly charges and 2,047 of CLTV. Support and risk analysis adds two useful signals: six of seven customers with recorded support activity churned, and the rule-based High Risk group captured all six observed churners.

Interpretation notes

Churn rates are within-group rates.

For example, 55.56% monthly churn means 5 of 9 monthly customers churned; these percentages do not represent each group's share of total churn.

Small-sample caution matters.

State-level and support-level results are based on only a few customers, so they should be treated as signals to investigate rather than stable population estimates.

Monthly charges are not absolute historical revenue loss.

The 73.94 figure is the sum of monthly charges for churned customers in this dataset. It is best described as monthly charges tied to churn rather than proven realized revenue loss over time.

Complaint metric needs careful wording.

The notebook's 0.33 value is the share of customers with recorded complaint data, not average complaints per customer. The dataset contains 9 recorded complaints across 21 customers, or 0.43 complaints per customer.

Association is not causation.

The escalation-vs-churn correlation of 0.47 is an observed association in a small recorded-support subset and does not establish that escalation causes churn.

The project demonstrates an end-to-end analytics workflow: relational data extraction, cleaning, feature engineering, exploratory analysis, visualization, and business-oriented interpretation. The strongest portfolio story is the ability to connect churn behavior with contract type, plan, acquisition source, support signals, and customer value.

Source: completed customer churn analysis notebook (main.ipynb). Figures and observations are drawn from the dataset and notebook calculations.